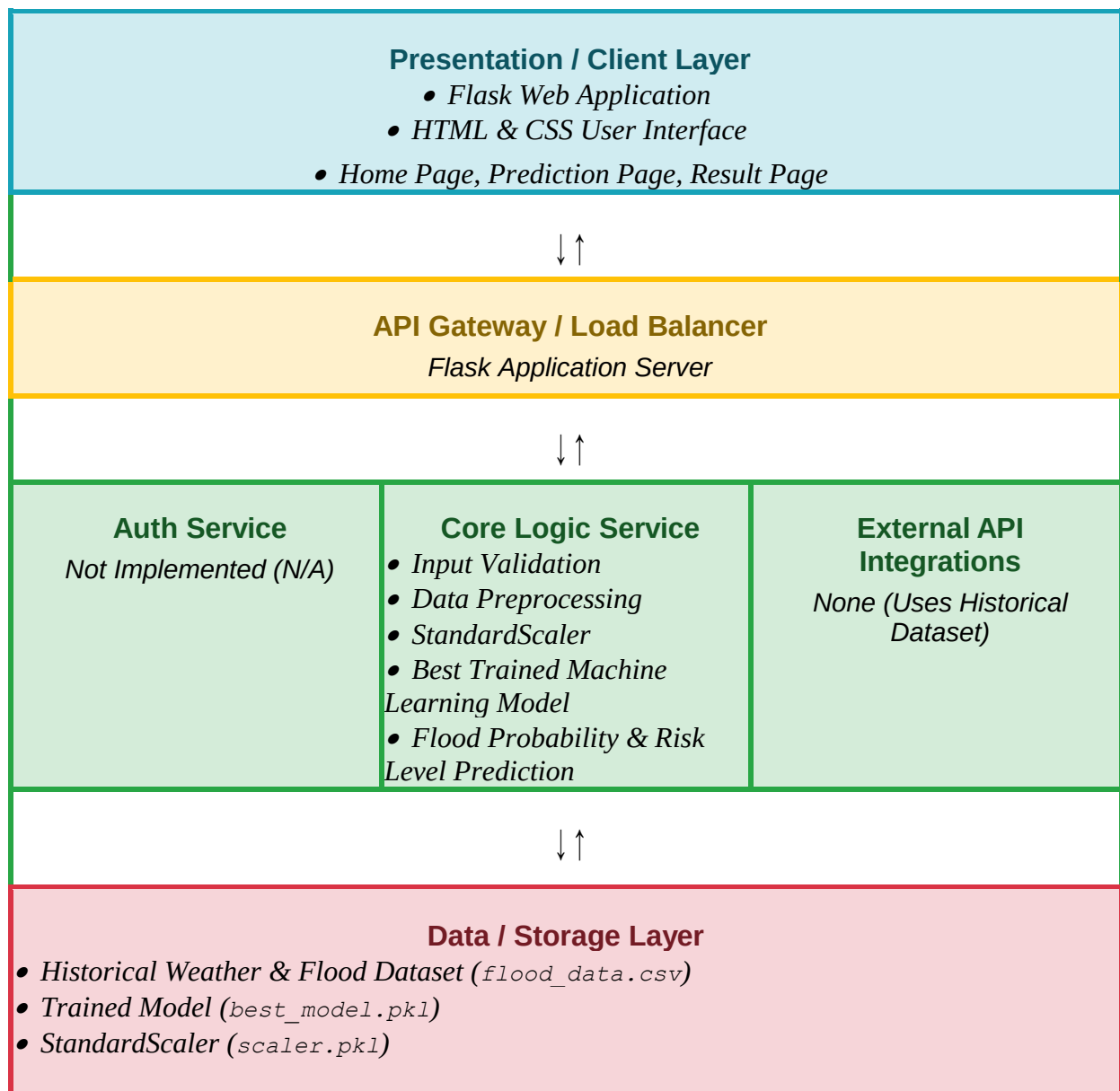


Solution Architecture

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides the user interface to enter weather data and display flood prediction results.</i>	<i>Flask, HTML, CSS</i>
<i>Application Layer</i>	<i>Processes user requests, validates input, performs feature scaling, loads the trained model, and generates flood predictions.</i>	<i>Python, Flask, Scikit-learn, XGBoost, Joblib</i>
<i>Core Prediction Service</i>	<i>Performs preprocessing and predicts flood probability using the best trained machine learning model.</i>	<i>Python, Scikit-learn, Joblib</i>
<i>Data Storage</i>	<i>Stores the historical weather dataset, trained machine learning model, scaler, and model metadata.</i>	<i>CSV (flood_data.csv), best_model.pkl, scaler.pkl, results.json</i>